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# Quiet Eye supports winner shots in a simulated table tennis competition

**Andrada Vincze**[1] | **Dragoș Iliescu**[1,2] | **Răzvan Jurchiș**[3]

[1]Faculty of Psychology and Educational Sciences, University of Bucharest, Bucharest, Romania

[2]Stellenbosch University, Stellenbosch, South Africa

[3]Cognitive Psychology Laboratory, Department of Psychology, Babeș-Bolyai University, Cluj-Napoca, Romania

**Correspondence**
Andrada Vincze, Faculty of Psychology and Educational Sciences, University of Bucharest, Bucharest, Romania.
Email: andrada.vincze@fpse.unibuc.ro

Răzvan Jurchiș, Cognitive Psychology Laboratory, Department of Psychology, Babeș-Bolyai University, Cluj-Napoca, Romania.
Email: razvan.jurchis@ubbcluj.ro

**Background:** QE is defined as the final fixation on a specific location or object in the visuomotor workspace, for a minimum of 100 ms, that enables the athletes to gather relevant information before a critical movement. Several lines of research indicate that QE contributes to sports performance. However, its contribution to performance is derived mainly from research investigating isolated motor tasks. Therefore, little is known about its contribution in realistic competitional settings.
**Methods:** The present study determined whether QE contributes to table tennis performance obtained from matches played in a simulated competition. Athletes ($N = 10$) performed two matches, one against a difficult and one against an accessible opponent. Gaze behavior was captured using Tobii Glasses 2.
**Results:** We found that athletes made longer QE before *winner balls* (i.e., balls that scored a point), compared to *forced* and *unforced errors* ($p_{bonferroni} = 0.005$, $t = -4.45$; $p_{bonferroni} < 0.001$, $t = -6.45$). Confirming that QE contributes to performance in a real match even in a competitional setting. We found no significant effect for the difficulty of the match, nor an interaction between the difficulty of the match and the type of shots $Fs\ (1, 9) < 2.26$, $ps > 0.16\ 0.70$, $\eta^2_{ps} < 0.20$.
**Conclusions:** The present study provides evidence for QE's contribution to sports performance investigated in a "*gold standard*" ecological environment. More specifically, it provides evidence that QE gradually increases with the quality of the shot. Identical patterns were found regardless of the difficulty of the match.

KEYWORDS
competitional setting, covert attention, elite performance, quiet eye

## 1 | INTRODUCTION

Attention focus, which is closely linked to performance in sports contexts, has traditionally been measured both in its overt and covert forms. Overt attention refers to a type of attention that is directly observable by researchers, usually based on the performer's eye movements (i.e., areas of interest and duration of gaze behavior, commonly measured with eye-tracking systems); in contrast, covert attention is measured through experimental methods that detect a mental shift that does not necessarily involve eye movements.[1] When investigating overt attention, the most robust type of gaze behavior that has been established to contribute to athletic expertise is Quiet Eye (QE).[2] QE is defined as the final fixation that is located on a specific location or object in the visuomotor workspace within 3° of visual angle for a minimum of 100 ms. The onset of QE starts prior to a critical movement (e.g., the backswing of a forehand stroke) and the offset manifests once the gaze deviates more than 3° from the fixated object.[2]

Robust evidence has been advanced for the contribution of QE to sports performance by four categories of findings: (1) Longer duration of QE is associated with greater athletic expertise;[3] (2) QE duration is longer in successful versus unsuccessful trials;[4] (3) stable QE durations in anxiety-provoking situations prevent athlete to choke;[5] and (4) training of QE leads to more effective performance enhancement compared to other types of training.[6] Finally, the contribution of QE to sports performance has been constantly replicated and confirmed by two meta-analytic studies.[7,8]

Diverse theoretical explanations have been proposed for this phenomenon, among which the information processing approach benefits from the strongest empirical support.[3] This perspective posits that athletes engage during the QE period in extensive cognitive processing in order to organize their subsequent action.[9,10] Hence, longer QE durations are considered to reflect an extended preparation period, which turns into higher chances of successfully executing a motor behavior.[11] Also, this explanation predicts that in more complex or more difficult tasks athletes need to align relevant parameters more precisely (e.g., spin and placement of the ball) in order to successfully execute a movement. The time needed to parametrize the tasks will increase consequently, which will result in longer QE durations. Therefore, it has been proposed that the performance-enhancing effects of longer QE are a result of effectively dealing with the complexity of the task,[12] but this hypothesis has received only sparse empirical investigation.[12–14] Evidence suggests that the relationship between QE and performance is not necessarily monotonous. In a study where complexity was manipulated threefold (i.e., the target size, the length of the golf putt, and the size of the effective putter), QE contributed to the management of only one type of complexity (i.e., length of the shot).[15] Therefore, the results regarding the pre-programming function of QE remain inconclusive. Our study extends the existing empirical evidence on QE by addressing three main limitations of the previous studies.

First, most extant studies on QE have focused on tasks that adopted trials consisting of isolated motor tasks (i.e., executing a single type of shot, like returning a serve in tennis), neglecting realistic tasks that require more dynamicity (e.g., a tennis exchange between two opponents). To our knowledge, there are no studies investigating realistic, dynamically performed sports, where athletes have to coordinate their bodies with an object that moves predictably or unpredictably toward them.[5] The best approximation we have so far comes from studying isolated motor tasks (e.g., receiving a service in a volley, goaltenders defending ice hockey shots, throwing a ball). However, these cannot be compared with sports where the main task of athletes is not limited to intercepting an object (e.g., the ball), but they also have to intercept and hit it on a very specific target and, even more, they have to perform a sequence of shots, not just a single shot, in order to increase the probability of success (e.g., tennis and table tennis). Aside from a previous study, where QE was not found to be a contributor to processing complex information in an interceptive sport that was dynamically performed, investigating QE in isolated motor tasks that require spatially reduced motor responses remained the norm.[16]

A second limitation of extant studies is that they investigated the contribution of QE to sports performance in *training-like* situations only. *Training-like* situations have indeed several advantages, being more similar to laboratory settings, where researchers have more control over irrelevant sources of variability, hence can ensure increased reliability of the measurements.[17] This is probably mainly due to imperfect eye-tracking systems that could be used in the early studies in QE and that were not fit for more ecologically valid scenarios; however, modern eye-tracking systems (i.e., those that have above 50 Hz sampling frequency) ensure the quality of the recorded gaze, even in a more dynamic, competition-like context, at least for some type of sports.[18]

Thirdly, since previous studies mainly used isolated shots, they were constrained to use a binary, dichotomic, coding of performance (including some of our previous studies[16,19]). On the contrary, when looking at the full progression of a match the binary coding of the outcomes (i.e., hits and misses) does not fully reflect a player's performance, because performance can be highly dependent on the opponent's game.[20] For instance, in an interacting sport, such as tennis, athletes have to constantly adapt to the opponents' strategic play, pace, velocity, and their performance will reflect how well the athlete is capable to adapt to the performance of their opponents. In tennis and table tennis, every point ends with one of the three outcomes: a "*winner*" when the player scores a point because the opponent is not able to reach the ball, indicating an excellent level of performance (e.g., the player placed the ball in an inaccessible location for the opponent); also, there can be two types of "*misses*": (1) "*forced errors*"—they are named *forced* because the athlete was out of balance or did not have enough time to react, despite their best efforts and (2) "*unforced errors*"—when the athlete was balanced, had enough time to strike accurately, but failed to do so. Unforced errors are commonly attributed to nervousness and are associated with higher levels of cortisol.[20,21] Forced errors, on the contrary, are frequently attributed to more aggressive styles of play that adopt an offensive position on the court.[22,23]

In order to better comprehend the relation between QE and performance, QE has to be investigated in dynamic tasks that also better simulate competitional environments. Further, for a more nuanced and realistic assessment of athletes' performance, we need to go beyond a dichotomic, hit versus missed, classification of performance, by considering the qualitative outcomes of the shots.

## 1.1 | The present study

The present study addressed the three abovementioned gaps. To address the first gap, we focus on an interceptive timing sport, namely table tennis. The structural features of table tennis (e.g., the fact that athletes gather relevant information only in front of them, commonly reduced (around 2 $m^2$) footwork) provide a favorable context to investigate the gaze behavior of the athlete while preserving a realistic, dynamic setting. Further, to address the second gap, we asked elite table tennis players to perform two real matches. Previous studies identified QE as a contributor to table tennis performance, but have adopted, as mentioned above, isolated motor tasks (i.e., using a single type of shot, statically performed).[24,25] The tasks previously employed differ significantly from realistic table tennis training exercises or matches, where athletes have more extended footwork and do not perform a predetermined sequence of strokes. In contrast, in the present study, the athletes played real matches. In order to provide these matches with a realistic stake, the athletes were announced by their club coach (that is also one of the national team coaches) that they will play in a tournament and that their results will be taken into consideration for the selection in the national team (see for a similar approach Vickers and Williams[5]).

Third, in order to assess performance in a realistic and nuanced manner, we compare the three types of shots: unforced errors, forced errors, and winners. In previous studies, which have used isolated shots and a dichotomic coding of performance (hit versus missed shots), qualitatively superior shots (hits) were preceded by longer QE durations.[4,7] Micro-level data suggest that momentary fluctuations (i.e., from point to point) in self-confidence influence the occurrence of unforced errors (i.e., when the levels of self-confidence are reduced) forced errors, and winners (i.e., when the levels of self-confidence are elevated).[26]

By employing a more granular assessment of performance (i.e., six gradually increased levels of performance), qualitatively superior volleyball shots were successfully predicted by working memory.[27] Following the abovementioned empirical findings regarding the requirement of an extended parametrization for a qualitatively superior shot and on the postulated role of QE in parametrizing the variables relevant for the subsequent stroke,[2] we expect QE to sustain qualitatively superior shots. Our *forte* is that we will empirically investigate differences in these types of shots and their associated QE in a realistic, dynamic, context of a match. Specifically, we expect *winners* to be preceded by longer QE, followed by *forced errors* (errors that are caused by a difficult ball), followed by *unforced errors* (errors that occur because of the poor preparation and/or execution of the athlete).

Formally, we hypothesize that:

**H1a** *In winner shots, athletes will display longer QE compared to forced errors and unforced errors.*

**H1b** *In forced errors, athletes will display longer QE compared to unforced errors.*

In order to test our second hypothesis, that the duration of QE extends as a function of effectively dealing with complexity, athletes performed two matches that differed in the level of complexity. Complexity was manipulated by varying how difficult the match was: In the low difficulty match (LD), athletes had a 70% chance to win the match, in the high difficulty (HD) match athletes 30% chance to win a match (as detailed in the Procedure). In the HD match, successful shots require more precise computations and calibrations from the athlete (e.g., of the velocity and the depth of their shot). A HD opponent has more advanced receiving techniques (e.g., backhand side-twisted), more powerful strokes, attacks more on the "weak side" (commonly the backhand) of the opponent, hence making the task more complex.[28] In accordance with the existing literature, we expect that participants will need to extend more their QE duration in order to score a qualitatively superior shot (versus an inferior one) in the HD match, as opposed to the LD match. In other words, we expect an interaction effect between shot quality (winners/forced/unforced errors) and the difficulty of

*In the preregistration form, all hypotheses were presented with reference to the trichotomous split of the quality of the shots (winners, forced, unforced errors), as in the manuscript. However, due to concerns regarding a possible lack of statistical power, in the preregistration we also specified the same hypotheses using a less detailed, dichotomic, split of shot quality (hit vs missed). This classification pools together forced and unforced errors in a single category (missed), which can increase statistical power. However, these analyses did not bring any additional data besides those obtained with the more nuanced classification (in winners, forced, unforced errors), that is, there was a significant effect of shot accuracy ($F = 38.52$, $p < 0.001$, $\eta^2_p = 0.81$), but no effects involving *match difficulty* ($Fs < 1.48$, $ps > 0.25$, $\eta^2_{ps} < 0.14$). Hence, they are not discussed in the manuscript.

the match (low/high) on QE duration. Specifically, we hypothesize that:

> **H2a** *In winners, as opposed to unforced errors, athletes will extend their QE durations more in the HD match compared to the LD match.*
> **H2b** *In winners, as opposed to forced errors, athletes will extend their QE durations more in the HD match compared to the LD match.*

## 2 | METHODS

### 2.1 | Participants

Following a previous study[16] and targeting a similar effect size (i.e., $\eta^2_p = 0.795$), we could achieve a power of 99% with only 3 participants (using G*Power 3).[29] However, given that it was likely that differences between conditions or interaction effects would have a lower magnitude, we aimed to gather as many participants as we could, while still keeping the elite level of the sample. Participants ($N = 10$) were *successful elite* (cf. Swan et al[30]) table tennis male players ($M_{age} = 15.02$ SD = 1.38). Since all the participants were underage, signed consent was obtained prior to participation from both the parents and the athletes. All participants had had a visual examination in the past six months and had normal or corrected to normal vision. Seven of them were right-handers and three of them were left-handers. We established two stringent inclusion criteria (1) table tennis experience above five years ($M_{experience} = 7.3$ years, SD = 0.88, *range* 6–8 years) and (2) at least one selection in the national team. The athletes were recruited from an excellence table tennis group, meaning that they were all among the top 20 players in their category in the country, and they play against each other several times in a season (see for more details, the supplementary material). Athletes have been monetarily reimbursed with $10 for their contribution to a larger research project (*see the author note*). The study was approved by the Research Ethics Committee of the University of Bucharest.

### 2.2 | Task

Each participant performed two table tennis matches as a part of a simulated competition. A table tennis match is usually played *the best out of five sets* (i.e., the first player to win three sets, wins the match), and each set is played up to eleven points. In the case of equality at ten points (i.e., 10–10), one player must score at least two consecutive points to win a set (e.g., 15–13). Between sets, athletes have a one-minute break.[31] The two matches performed by the athletes differed in their level of difficulty. In the low difficulty match (LD), athletes had a 70% chance to win the match, in the high difficulty (HD) match, athletes had a 30% chance to win a match. The level of difficulty was established based on head-to-head encounters coming from previous competitions. The order of the two matches was counterbalanced with the aim of preventing order effects to affect our results.

### 2.3 | Eye Tracking Equipment

We utilized a pair of *Tobii Glasses 2* to capture the participants' gaze behavior. The glasses use a pupil-centered corneal reflection, dark pupil eye-tracking technique, binocular, using 4 eye cameras that capture video at a rate of 50 Hz (Tobii Pro AB). The glasses are connected by a wire to the recording unit which was attached to the athletes' shorts and affords the athlete to perform a table tennis match in a near to normal state. Prior to the debut of each match, the researcher (the first author) calibrated the glasses using a one-point calibration procedure provided by the Tobii Software; in addition, athletes were also requested to look at other three-four targets, to ensure the tracking precision (as recommended by Holmqvist et al,[32]). We used the Tobii Pro Lab Software (version 1.1.5.2) to create and analyze the eye-tracking data. The Software automatically provides the type (e.g., fixation and saccade) and the duration of gaze behavior. Each fixation is marked on the video with a red dot, therefore allowing us to detect the location of the fixation and to conduct extended analyses based on the recorded scene.

### 2.4 | Procedure

After data collection, but before analyzing or looking at any of the data, we preregistered all hypotheses and statistical analyses here: https://osf.io/7b4mh/. We followed the protocol of van den Akker et al, for preregistration after data collection.[33] All sports facilities (e.g., rackets, balls, and tables) complied with the standards of the *International Table Tennis Federation*. The coach had a short discussion with the athletes, where he announced that everyone will play two matches while connected to the eye-tracking system and that those matches will be considered for their next national team selection. Since the club coach is also one of the national team coaches and the athletes were already part of the extended national team, the manipulation was highly credible. Each athlete executed their physical warm-up that was identical in duration (i.e., 20 min) and content (i.e., active, passive, and flexibility warm-up) and after that, they were connected to the eye-tracker. Athletes were informed that if they feel that the eye tracker moves (e.g., slipping because of sweat) or if they feel any disturbances related to the eye-tracking system, they could request a short break to fix it. Since they had a previous experience with the eye-tracker (in the past 6 months), no adaptation time was provided anymore. The athletes performed two matches in a counterbalanced order (high and low difficulty). Before the start of each match, the athletes rated their competitive anxiety, measured with Mental Readiness Form-3 (see Measurements). Performing two matches in a day is below what a typical competition day is in table tennis competitions, where athletes perform up to 4 matches in a day (cf. National Federation Site). Therefore, fatigue could less likely affect our results. Whenever the servers changed (after 2 points), the score was announced out loud by the player that was analyzed. They had 1-min break after each set and a five-minute break between the two matches. After the second match participants were thanked and debriefed.

## 2.5 | Measurements

### 2.5.1 | Quiet Eye

The Quiet Eye was previously defined as a fixation that lasts for at least 100 ms that occurs in the final moment before the athlete initiates a movement (i.e., a backswing in table tennis). For table tennis, the location where athletes are fixating is on the ball, in the exact moment when the opponent touches the ball with their racquet.[16,19] The cut-off of 100 ms was established according to the capacity of the eye-tracking systems previously utilized that were able to update the gaze behavior once every 33.3 s and needed at least three frames to detect a fixation (i.e., resulting in a shortest minimum fixation of 100 ms).[2] Modern eye-tracking systems, as it is Tobii Glasses 2, update the position of the gaze behavior every 20 ms and it detects a fixation when the gaze is stable for at least 4 frames, allowing, hence, the detection of fixation of 80 ms. Consequently, we have also analyzed trials that had the shortest fixation at 80 ms. Two types of trials were eliminated; those that were performed without a final fixation due to technological errors and those that while playing touched the let or the edge of the table, making the ball almost impossible to be played. Therefore, we have eliminated 8.89% of the overall performed balls. Even with the eliminated trials, we analyzed on average above 100 trials for each participant ($M_{analyzedtrials} = 102$, $SD = 37.84$). Due to the fast pace and unpredictability of the task, it was not possible to use any automatic coding functions of the software. Thus, the correspondence between the gaze behavior and performance (i.e., which fixation corresponded to which ball and whether the ball was hit or missed) was conducted by the first author (that has also a table tennis experience above 20 years) in a frame-by-frame approach. Furthermore, the eye tracker does not record just the gaze behavior of the athlete but also what the athlete sees in the environment while performing the task. For each analyzed shot, we took the fixation that occurred the closest after the *opponent* touched the ball. Only after this moment, the athlete could initiate the preparation of the movement because only after this moment the athlete could detect information that would help them to prepare for their next stroke. A second coder (the third author) reanalyzed 115 points from two randomly selected matches, including one in the low difficulty and one in the high difficulty condition. The re-analyzed points represent more than 10% of the total number of points included in the study (i.e., a total of 1020 points; see for a similar approach Wood et al,[34]). Identically to the first coder, the second one identified, for each point, which was the final fixation that preceded the initiation of the stroke that resulted in winning or conceding the point. The results showed an excellent level of agreement: namely, for 95.56% of the points, the second rater identified as "the final fixation" the same fixation as the first coder.

### 2.5.2 | Table tennis performance

Table tennis performance was coded by two experienced table tennis players (i.e., that have table tennis experience above 10 years and have been both members of the national team) and had an *almost perfect* agreement ($k = 0.88$, CI = [0.84–0.92]).[35] Videos exported from Tobii Pro Lab were slowed down at one eight (1/8) of the original speed. At this speed, the different types of shots were clearly distinguishable. Table tennis points are commonly played out of at least two strokes (i.e., service and receive). However, there are instances wherein to score a point, an athlete has to strike more than three shots. In all the points where the athletes performed more than three shots, the raters coded the qualitative outcome of the final shot, that could be (1) an unforced error—when the athlete made a mistake that is attributed to one's failure, (2) a forced error—when the athlete is forced by the opponent to make an error (e.g., by a strategic placement of the ball) and (3) a winner—when the player is awarded with one point. The raters also coded the performance at a macro level, whether the athlete won or lost a match, and at a macro level, there were no disagreements.

### 2.5.3 | Mental Readiness Form (MRF-3)

MRF-3 is a scale that measures competitive anxiety, by assessing cognitive anxiety (i.e., negative expectations or concerns about performance), somatic anxiety (i.e., tension as physical manifestation), and confidence levels (i.e., positive expectations regarding their upcoming performance). Each concept is rated separately on an 11-point anchored *Likert*-type: "worried-not worried," "tense-not tense," and "confident-not confident." Higher scores for cognitive and somatic anxiety indicate higher competitive anxiety. MRF-3 has shown high concurrent validity with other anxiety measurements (e.g., CSAI-2).[36] We use MRF-3 for assessing whether participants evaluated the HD match as being indeed more difficult and threatening compared to the LD match.

## 3 | RESULTS

### 3.1 | Manipulation check

As a means to test whether our complexity level for the matches was appropriately set, we evaluated whether the number of won matches was lower in the HD condition. In accordance with our expectation, a McNemar test indicates that the number of won matches was lower in the HD condition $\chi^2 = (1, 10) = 5.000$, $p = 0.025$. A proportion of 80% of the athletes won their LD match and just 30% of the athletes won their match in the HD condition.

Furthermore, we reasoned that higher competitive anxiety would be an indirect evidence for our manipulation of the level of difficulty. Paired *t*-test indicate that cognitive ($t\,(9) = 5.29$, $p < 0.001$, $d = 1.67$) and somatic ($t\,(9) = 3.64$, $p = 0.005$, $d = 1.15$) anxiety were higher before the HD compared to LD matches, and the levels of confidence were lower ($t\,(9) = -4.18$, $p = 0.002$, $d = -1.32$) before starting the HD match compared to when they started their match in the LD condition. Putting together these results, we have robust arguments to conclude that our complexity manipulation was a fruitful one, and athletes had higher difficulties in winning a match when they performed in the HD condition.

### 3.2 | Descriptive statistics

Table 1 presents for informative purposes the descriptive results of QE in milliseconds split on type of difficulty of the match (low and high) and of performance (winner, forced error, unforced error).

### 3.3 | Inferential statistics

We conducted a 2×3 within-subjects ANOVA to test for possible differences at the two levels of difficulty (low, high) and the quality of the shots' outcome (unforced errors, forced errors, winners). Regarding our first hypothesis, we found a significant effect for the type of shot $F\,(1, 9) = 24.62$, $p < 0.001$, $\eta^2_p = 0.73$. Bonferroni post hoc tests indicate that forced errors were preceded by longer QE compared to unforced errors ($M_{difference} = 147$ ms, $SE = 42.8$, $t\,(9) = 3.43$, $p_{bonferroni} = 0.002$). Moreover, the post hoc tests indicate that winner shots were preceded by longer QE compared to forced errors ($M_{difference} = 107$ ms, $SE = 24$, $t\,(9) = 4.45$, $p_{bonferroni} = 0.005$). These results indicate that QE duration increased gradually with the quality of the shots' outcome (*see* Figure 1).

We found no significant effect for the difficulty of the match $F\,(1, 9) = 2.26$, $p = 0.16$, $\eta^2_p = 0.20$, indicating that differences in the duration of QE were not found across the quality of the shot's outcome when performed in low difficulty match (*estimated marginal mean* = 699.97, SE = 32.15) or in a high difficulty match (*estimated*

**TABLE 1** Descriptive results of Quiet Eye in milliseconds split on type of match and shot outcome

| Difficulty of the match | Performance | Mean QE | SE | 95% confidence interval Lower | Upper |
|---|---|---|---|---|---|
| Low difficulty match | Winner | 811.05 | 39.01 | 722.80 | 899.30 |
| | Forced Error | 729.87 | 37.44 | 645.17 | 814.57 |
| | Unforced Error | 559.00 | 56.61 | 430.94 | 687.07 |
| High difficulty match | Winner | 785.34 | 47.33 | 678.29 | 892.40 |
| | Forced Error | 653.15 | 35.54 | 572.76 | 733.54 |
| | Unforced Error | 530.04 | 39.27 | 441.20 | 618.87 |

*Note*: For each type of shot, the mean QE reflects an average score for that specific shot.

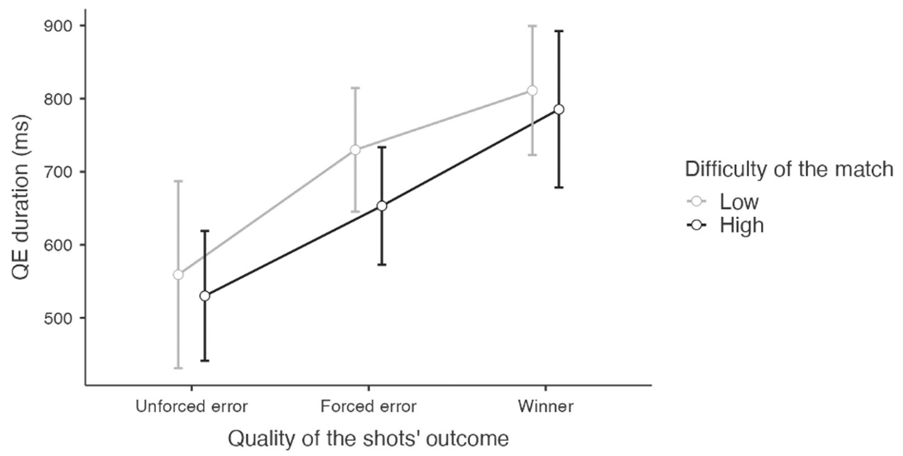


**FIGURE 1** Quality of the shots' outcome split on the difficulty of the match. Error bars represent a 95% confidence interval (CI) of the mean duration of QE.

*marginal mean* = 656.17, SE = 35.34). When testing the second hypothesis, we did not find a significant interaction between the difficulty of the match and the quality of the shots' outcome $F(1, 9) = 0.48$, $p = 0.62$, $\eta^2_p = 0.051$.

## 4 | DISCUSSION

In this study, we investigated whether QE plays a role in the performance and in the quality of the shots in a fully performed sport, during real matches from a simulated competition. We also evaluated whether QE works as a mechanism to effectively deal with complexity (manipulated by varying the difficulty of the match).

We first found that QE sustains table tennis performance, given the fact that the duration of QE increased linearly with the quality of the shots: *winners* were preceded by longer QE compared to forced errors, which, in turn, were preceded by longer QE than unforced errors. Importantly, this constitutes one of the first evidences that QE contributes to performance in a realistic match from a simulated competition where athletes perform against an opponent. Starting from the assumption that winner shots require the most precise parametrization (i.e., managing more precisely variables such as placement of the ball, spin, and timing) the time needed to efficiently parametrize is reflected in longer QE durations. In the same vein, longer QE durations before forced errors, compared to unforced errors indicates that in those mistakes that are forced by the opponent, the athletes engage, at least in part, in some cognitive process in order to respond to the game situation.

Second, regarding the complexity hypothesis (i.e., that one of the functions of the QE is to effectively deal with complexity) we investigated whether in the more difficult match (i.e., where the requirements of the task were more complex) athletes would display longer QE, indicating that an increased level of cognitive processing aimed at dealing with the increased complexity. Inconsistent with the previous literature[12–14] that investigated this function, we found no evidence to support our hypothesis. We cannot firmly conclude, based on a non-significant result obtained on a small sample, that QE does not act as a mechanism to deal with complexity. The non-significant result could indicate an absent effect of complexity but could also indicate low statistical power. However, one could make a case for the null effects of complexity, and it comes from the way athletes approached the difficult match. In seven out of 10 instances, the athletes lost their difficult match (i.e., where athletes had just 30% chances to win the match). The overall number of won matches across participants indicates that a priori levels of difficulty were appropriately set. Moreover, in this match, the athletes were able to score very few points. This means that on average athletes were not able to score more than seven points in a set and in some sets, they lost by an extreme difference of points (e.g., 11–2).

The *attentional control theory* (ACT) might provide an explanation for the fact that, although we succeeded in manipulating the level of task complexity, QE did not increase with complexity. ACT's core postulate is that when worries interfere with the task at hand, it consumes the processing resources and therefore the performance can be affected.[37]

The fact that we found higher levels of anxiety (including *cognitive* anxiety, which indexes worry) shows that worry was present, and, according to ACT, it could have disrupted the athletes' visual attention resources, which then translated into an impaired ability to properly increase the duration of QE.

### 4.1 | Strengths and limitations

The most important strength of our study is that we extended the investigations on the contribution of QE to performance, to more ecologic and dynamic environments, such as a simulated competition for an interacting sport, where athletes have to constantly adapt to an opponent. Furthermore, we also grow beyond the binary approach, where performance is artificially split into "hits" and "misses," by having a more qualitative approach regarding the outcome of the shots. Nonetheless, we would readily admit the reality is probably even more complex than that, and future research could have a more granulated approach when investigating winner shots (e.g., a winner shot because of a good tactical decision, or a winner shot because of an unforced error of the opponent).

One possible limitation of the study is that we lost somewhat more trials (8.89%) compared to other studies where the motor response was spatially reduced (e.g., 6.1% Klostermann et al,[38] but also less compared to others, see Harris et al[10]). Due to the constant movement and the fast pace of the sport, the number of trials where the image was not available from either of the eyes increased. Note, however, that we analyzed a comprehensive number of trials for each participant, therefore allowing us to proceed with the pre-planned statistical analyses and to detect significant effects that support our first hypothesis. Another possible limitation of the study is that we did not have one more match, where we could test whether a moderate level of difficulty would affect less the emotional state of the athletes, allowing us to eventually detect whether the function of effectively dealing with complexity is supported also in more ecological environments (i.e., a simulated competition).

### 4.2 | Perspective

Along with several other studies (see for a summarization Lebeau et al[7]), we found that Quiet Eye is a robust contributor to sports performance. Physical and other mental skill training have exponentially evolved in the last decade; contrastingly, even with all the available modern technologies, few resources are devoted to the training of QE and to other similar types of skills that have been shown to highly contribute to performance (e.g., anticipation). Integrating QE training into regular training programs could enhance the acquisition of new skills, refine skills that are already mastered, or could contribute to a more resilient response under anxiety-provoking situations (Vine et al[6]). By adopting a more ecological environment, we only had a noble goal, to bring closer the research and the applied field, and from now on conceivably to positively contribute to numerous lives of elite athletes.

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## CONFLICT OF INTEREST

None.

## AUTHOR NOTE

The present study is part of the Ph.D. thesis of the first author that, in a series of studies, aims to investigate the role of Quiet Eye (QE) in elite table tennis performance, and, also, the influences on QE of several factors: task complexity, difficulty, fatigue, and stress. The studies share some methodological commonalities: They are conducted on the same elite sample of table tennis players and use a similar apparatus (e.g., measure QE using the same eye-tracking system). However, importantly, the different studies test different hypotheses using different data sets, collected from different experiments and tasks.

## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author, Vincze Andrada. The data are not publicly available because they were gathered from athletes that can be identified (i.e., based on their national and international official rankings); hence, their privacy would be affected. The study was approved by the Ethical Committee of Research of the University of Bucharest with the decision number 98/05.12.2019.

## ORCID

*Andrada Vincze* https://orcid.org/0000-0001-6169-6214

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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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